

Taming the Beast

Secure Retrieval-Augmented Generation for T&E Report Automation in Classified Environments

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Why RAG? The Problem with Standalone LLMs

Three Critical Failure Modes

- **Knowledge staleness** – test procedures, MIL-STDs, and specs evolve continuously; LLM weights are frozen at training time
- **Hallucination** – models generate plausible but incorrect outputs in safety-critical reporting
- **Non-traceability** – outputs cannot be traced to authoritative sources, violating DoW documentation standards

Fine-Tuning Is Not the Answer

- High retraining compute costs, must repeat with new data
- Degrades performance outside the fine-tuned domain
- No source attribution for auditability
- Increased vendor lock-in with a single LLM

RAG combines a librarian with a language model: the librarian searches, the model writes.

How RAG Works

The Four-Stage Pipeline

1. **Document Ingestion** – Parse, normalize, and chunk T&E artifacts (test plans, MIL-STDs, specs, reports)
2. **Embedding & Indexing** – Encode chunks into dense vectors; store with metadata (classification, date, system ID)
3. **Query-Time Retrieval** – Embed the user's query; find top-*k* most similar chunks via cosine similarity
4. **Augmented Generation** – Inject retrieved chunks into the LLM context; generate a grounded, cited response

Key Insight: Semantic Search

Embeddings encode **meaning**, not keywords:

- *"the dog chased the ball"*
- *"a puppy ran after a sphere"*

These map to **nearby points** in vector space – enabling retrieval that keyword search cannot achieve.

RAG accuracy: **107-177% improvement** over zero-shot LLM baselines on domain QA tasks

Naive vs. Advanced RAG

Naive RAG

Single-pass retrieve-then-generate

- Prone to low-precision retrieval
- Context misalignment
- No quality filtering

Advanced RAG

Multi-stage processing pipeline

- Query rewriting & expansion
- Re-ranking (3.4x precision improvement)
- Post-retrieval compression
- Relevance filtering

Agentic RAG

LLM orchestrator decides autonomously

- When retrieval is needed
- Multi-hop sub-queries
- Iterative retrieval rounds
- Already in DoD AI RCC pilots

Production T&E deployments require **Advanced RAG** at minimum – naive RAG is not sufficient

Chunking & Hybrid Search: Getting Retrieval Right

Chunking Strategy Matters

Chunk Size	Faithfulness	Relevancy
256 tokens	95.56%	97.78%
512 tokens	97.59%	96.67%
2048 tokens	80.37%	91.11%

- **Sliding window** with overlap outperforms fixed chunking
- **Semantic-boundary** chunking
(paragraph/section) preferred for structured T&E documents

Hybrid Search

Combines dense vector search + sparse BM25 keyword search:

$$S_h = \alpha \cdot S_s + S_d$$

At optimal $\alpha = 0.3$:

- **mAP: 47.14** (vs. 30.13 BM25 alone)
- **nDCG@10: 72.50**

Essential for T&E terminology: MIL-STD designators, system nomenclature, test procedure IDs

Production Architecture: Five Layers

L1: Ingest

SharePoint, SIPR file shares, PDFs, TDPs

OCR, table extraction

Metadata enrichment

Deduplication

L2: Index

Semantic chunking (256-1024 tokens)

Embedding models (cloud or air-gapped)

Vector DB: Milvus, Weaviate, pgvector

Hybrid search

L3: Retrieve

Query expansion & HyDE

Re-ranking

Metadata filtering (classification)

Context formatting

L4: Generate

IL4: GPT-4o, Gemini, Claude

IL5: Ask Sage (FedRAMP High)

IL6: Dedicated SIPRNet enclaves

Prompt engineering

L5: Govern

Immutable audit trail

RAGAS metrics

AI CI/CD pipeline

Human-in-the-loop review

A production DoW RAG system is a **multi-layer data platform** – not an ad hoc chatbot

Deployment Across Classification Levels

JWCC Cloud On-Ramps

Four enterprise providers – AWS, Azure, Google, Oracle – supporting IL2 through IL6

Level	Environment	Requirements
IL4	NIPRNet	Multi-tenant, CAC auth, FIPS 140-2
IL5	NIPRNet Sensitive	Dedicated enclave, US-citizen personnel, 421+ NIST controls
IL6	SIPRNet	SECRET clearances, TEMPEST shielding, air-gapped inference
DDIL	Tactical Edge	Containerized RAG on vessels, FOBs

GenAI.mil: The Mandate

- **Dec 2025:** CDAO launches GenAI.mil – Pentagon-wide generative AI platform
- **Jan 2026:** DoN mandates transition for all commands; **deadline April 30, 2026**
- July 2025 CDAO frontier-AI contracts, up to \$200M each: **Google, OpenAI, xAI** (*Anthropic excluded since March 2026*)
- Gemini for Government first; RAG with uploaded docs, web grounding, deep research
- **3 million** warfighters, civilians, and contractors served

RAG is now an **institutional requirement**, not an optional capability

Security Controls for CUI and Classified RAG

Regulatory Framework

- **DoDI 8510.01** – RMF for all DoW IT (including RAG)
- **NIST SP 800-53 Rev 5** – Control catalog (AC, AU, SC, SI, RA families)
- **NIST SP 800-171 Rev 3 / CMMC L2-3** – Contractors processing CUI
- **NIST AI RMF 1.0 + AI 600-1** – GenAI-specific guidance
- **DoD Zero Trust Strategy** – 152 ZT activities across 5 phases

RAG-Specific Controls

- **Classification-aware ingestion** – Parse and store markings as vector metadata; separate indexes by level
- **ABAC at query time** – Users only retrieve chunks at/below their clearance
- **Encryption** – AES-256 at rest (FIPS 140-2), TLS 1.3 in transit, mTLS service-to-service
- **Embedding confidentiality** – Dedicated inference nodes; vector similarity attacks are real
- **CAC/PIV authentication** – All RAG endpoints

Zero Trust: every identity continuously verified, no implicit trust for in-network components

AI-Specific Threats to RAG Systems

Knowledge Base Poisoning

PoisonedRAG attack: **5 malicious texts** in a **2.68M**-document corpus achieved a **97% attack success rate** – paraphrasing and perplexity detection proved **insufficient**

Mitigations:

- Provenance verification
- Integrity hashing
- Access-controlled ingest

Prompt Injection

Adversarial content in retrieved documents instructs the LLM to override system prompts

Mitigations:

- Input sanitization at ingestion
- Prompt shields
- Output validation

Model Inversion & Hallucination

- Embedding queries can leak information
- Hallucinations persist even with good context

Mitigations:

- Audit logs on all queries
- Groundedness scoring
- Mandatory human review

Defenses must be **layered**: provenance verification, integrity hashing, access-controlled ingest, adversarial audits

The Anthropic Lesson: Why Vendor Independence Matters

What Happened (Feb–Mar 2026)

- July 2025: Claude approved for classified networks
- Feb 27, 2026: President directed all agencies to cease using Anthropic
- Mar 5, 2026: DoW issued supply chain risk designation
- GSA removed Anthropic from USAi.gov
- Navy commands began removing Claude models

The Architecture Lesson

Organizations locked to a single LLM provider faced **immediate operational disruption**

Systems with **abstraction layers** transitioned to GPT-4o, Gemini, or open-source alternatives with **minimal changes**

Treat the LLM as a **replaceable component** behind a standardized interface

Frameworks like LangChain and Haystack enable vendor-agnostic design: swap a model identifier, not the pipeline

Case Study: RustyAI – Cloning Navy's Corrosion SME

The Mission

A senior Navy corrosion engineer is approaching retirement – decades of specialized knowledge with no succession plan.

Goal: capture and replicate that expertise before it walks out the door.

~30 GB | ~20,000 Files

Emails · technical handbooks · presentations · engineering reports

Outcome: AI assistant that answers in the SME's **own style**, with their depth of expertise.

The Platform: RustyAI

Four-tier deployment on AWS GovCloud (us-gov-west-1):

Frontend	Streamlit + Nginx (TLS 1.3)
Application	Haystack RAG framework
Data	Milvus + MinIO + Etcd
Inference	Amazon Bedrock (Titan + Claude)

Ingestion: Docling (OCR + VLMs) on NVIDIA L4 – 2 hr → 20 min;
Markdown chunks → Titan embeddings → Milvus

XML-delimited prompts w citation require. for DoW traceability

RustyAI: Lessons Learned

- 1. GPU acceleration is not optional at scale** – the order-of-magnitude speedup determines whether corpus updates complete within operationally acceptable windows
- 2. Right-size your vector DB** – Milvus Lite (Python library, t3.small) scales to millions of vectors; Standalone (t3.xlarge) scales to 100M+ for enterprise workloads
- 3. TLS everywhere, even internal** – GovCloud environments block unencrypted traffic and non-standard ports
- 4. Modular frameworks pay off** – Haystack's component architecture allowed individual pipeline stages to be replaced without affecting the overall system
- 5. Data conversion is key** – scanned PDFs of handbooks, Excel spreadsheets, and videos are all similarly converted to a structured format before ingestion

Few-Shot Prompting: From Q&A Tool to Report Generator

The Conceptual Shift

A baseline RAG system **answers questions**

A few-shot-prompted RAG system **produces structured deliverables**

The difference is entirely in the **prompt engineering layer** – retrieval and generation architecture unchanged

PPV improvement: 15-25 percentage points from zero-shot to few-shot

Prompt Architecture

1. **System prompt** – Role, output constraints, chain-of-thought
2. **Few-shot block** – 3-5 complete example report sections with annotated context
3. **Query block** – Analyst's request
4. **Context window** – Top- k retrieved chunks

Retrieved context provides **factual grounding**; few-shot examples specify **format and reasoning style**

T&E Report Types Amenable to RAG Automation

Templated Reports

- **TEMP sections** – Background, test objectives, resource requirements
- **Test Execution Reports** – Structured findings with condition/deviation/severity
- **Test Incident Reports** – Deficiency with corrective action fields
- **MOP/MOE dashboards** – Metrics tables from raw test data

Multi-Document Synthesis

- **Interim Fielding Assessments** – Operational effectiveness/suitability across multiple test events
- **Deficiency Reports / FRACAS** – Condition/cause/corrective action from historical databases
- **After-Action Reviews** – Synthesize across hundreds of T&E artifacts simultaneously

95% savings on ATO documentation (Ask Sage)

60-80% reduction in initial draft time

100K+ users on DoW GenAI platforms

*Analyst role shifts from **drafter** to **reviewer** – Test for Less*

Evaluation Framework

Automated Metrics

RAGAS Framework:

- Context relevance
- Answer faithfulness
- Answer relevance

ARES Framework:

- Few-shot-seeded LLM judges
- Validated against human preference annotations

T&E-Specific Metrics

- **Citation coverage rate** – Are all claims traceable?
- **Format compliance rate** – Does output match the template?
- **Deficiency detection rate** – Compared to ground-truth test records

Before Deployment

- Red-team evaluation for prompt injection, poisoning, and exfiltration
- Regression testing on fixed evaluation sets at each pipeline update
- Human-in-the-loop review **mandatory** for all T&E deliverables

Closing: RAG Is Ready for T&E

The Capability

107-177% accuracy
gain Multi-document
synthesis at scale
Source attribution
built in

The Mandate

GenAI.mil required by
April 30 CDAO AI RCC
sprint methodology
DoD Zero Trust
integration

The Path Forward

Pilot on NIPRNet IL4
(90-day eval) Security
review for IL5/IL6 ATO
Few-shot template
library for top 5 T&E
deliverables

RAG is the highest-value, lowest-risk AI capability for T&E today

It does not require retraining, does not export classified data to vendors, and deploys incrementally.

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